الشركة العُمانية لنقل الكهرباء ش.م.ع.م
OMAN ELECTRICITY TRANSMISSION COMPANY S.A.O.C
إحدى شركات مجموعة نماء
Member of Nama Group

Reference: OETC/CTO/2025/ **871**
Date: **8** October 2025

Ghalib Al Mamari
Chief Executive Officer (Acting)
OQ Alternative Energy

After Compliments,

Subject: Connection Offer for Marsa Solar

With reference to the above subject and OQAE Connection Application dated 11th May 2025, ref: OETC/NP/CA/2025/20, requesting to connect Marsa Solar, with a maximum export capacity of 280 MW, to the Transmission System, OETC would like to thank you for submitting the Connection Application.

This letter contains OETC's offer for connecting Marsa Solar to the 400 kV transmission system. According to the Connection and Use of System Charge Methodology Statement (approved by the Authority), clause-3 Connection Charges, there are Connection Charges to be paid by the applicant. The Connection Charges, according to clause 3.3 of the Connection and Use of System Charge Methodology Statement, include two components:

1. Capital Charge
2. Transmission Running Charge (Operation & Maintenance Charges)

<u>Capital Charge:</u>

The total estimated cost of Connection Assets for the provision of 2x400 kV GIS bays, civil works, auxiliary power, etc., to be paid is initially calculated as a total sum of RO 3,290,400 (Omani Rial three million two hundred ninety thousand four hundred only).

Transmission Running (Operation and Maintenance) Charge:

The Transmission Running (Operation and Maintenance Charge) to be payable annually will be based on the Transmission Running Charge (TRC) factor, calculated according to item 3.3.3 Transmission Running Charge of the Connection and Use of System Charge Methodology Statement. As an indicative value only, the annual TRC percentage is expected to be between 1% to 3% of the GAV of Connection Assets (including CPI adjustments).

Please consider the following conditions for Marsa Solar acceptance:

- The developer must ensure compliance with Grid Code requirements at the connection point and install dynamic reactive compensation devices and Flexible AC Transmission System (FACTS) devices to provide necessary voltage support, as validated through Grid Impact Studies.

- Marsa Solar should be equipped with grid-forming inverters capable of providing voltage and frequency control, inertial response, and fault current contribution to enhance grid stability and support system restoration.

- Marsa Solar shall incorporate a black start facility in accordance with Transmission System requirements. Therefore, a Battery Energy Storage System (BESS) shall be installed to mitigate Solar production intermittency and maintain system stability at the connection. The sizing of the storage will be determined as part of the Grid Impact Studies.

- The developer is to schedule the energization activities during the maintenance period only (1st of October to 31st of March of the following year). No energization, testing, and commissioning activities will be allowed during the restricted period (1st of April to 30th of September) as per Transmission Security Standard.

- The connection charges are only estimated values at this stage. Final values will be updated in the Electrical Connection Agreement (ECA).

- The mentioned connection charges are for the up-front payment option.

- As per the connection methodology, the customer should accept this offer within five weeks; otherwise, the offer will lapse.

- Considering the design and construction time required, we would like to confirm that the connection points for Marsa Solar will be ready by Q2 2028.

- For the calculation of Annual Connection Charges (Capital portion and O&M portion), the asset values will be indexed each year by the Consumer Price Index (in accordance with the Connection and Use of System Charging Methodology Statement applicable at the time).

- Application Fees paid by the System User will be deducted from the Connection Charges in accordance with the Connection & Use of System Charging Methodology.

- OETC assumed that the auxiliary supply LVAC of the grid station would be provided from the developer side, and in addition, via an 11 kV connection from the nearest primary substation (which belongs to NEDC).

- If the application is cancelled by the applicant (OQAE) after the acceptance of the connection offer, or in case of delay caused by the applicant to implement the connection to the Transmission System for a long period, the applicant shall indemnify the Licensed Transmission System Operator (OETC) from and against all costs and expenses necessarily incurred by OETC in undertaking works in respect of the acceptance of the Connection Offer provided by the applicant.

- For a claim under the previous Item, OETC may provide supporting evidence (including purchase order details and supplier invoices) of costs and expenses incurred by OETC in undertaking such work as may be reasonably required.

- As per the Connection and Use of System Charge Methodology Statement (clause 3.3.1), the applicant shall be liable to pay interest during construction, which is 7.61% of Gross Asset Value (as per approved Price Control-6). This value is estimated and will be revised based on the actual capex spent and the project duration, which will be updated in the Electrical Connection Agreement (ECA).

- OETC Quality, Health, Safety, Security, and Environment (QHSSE) requirements inside the grid station shall be considered during the different phases of your project. Please refer to the Grid Code requirement (OC6 Safety Coordination) for safety document exchanges and to fulfill all requirements.

- The following required detailed system studies (but not limited to) shall be submitted by the developer and approved by OETC:
  - **Grid impact study**
    - Healthy State (N) Voltage Profile
    - Power Flow Intermittency Analysis
    - Contingency Analysis
    - Short Circuit Analysis
    - Transient Stability Analysis
  - **Grid Code Compliance Study**
    - Connection Assessment
    - Load Flow Studies
      - Power Factor Control Assessment
      - Reactive/Voltage Control Assessment
      - Power Factor/Reactive
    - Fault Ride Through
    - Frequency Response
    - Reactive Power Response
    - Power Quality
      - Voltage Unbalance
      - Flicker Study
      - Frequency Scan & Harmonic Distortion (HD)

We are looking forward to receiving your confirmation of acceptance of this offer and the above conditions. Please do not hesitate to contact us for any further discussion or information.

Yours faithfully,

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Ding Maosheng
Chief Technology Officer

Cc:
- Chief Executive Officer, OETC.
- General Manager of the Planning and Design Department, OETC.
- File.

Table (1) Connection Charge for OQAE (Site Sadaf GS)							
Assets used to connect OQAE to OETC System at Sadaf Grid Station							
Sr. No.	**Asset**	**Type of Asset**	**Qty**	**Total Cost (R.O)** **A**	**Cost for OETC (R.O)** **B**	**Cost for User (R.O)** **C**	**Remarks**
1	400 kV GIS with Control & Protection Panels with Bus Duct between 400 kV OHL & GIS.	Transmission	17 nos.	13,391,430	13,391,430	-	The full amount is to be paid by OETC.
2	400 kV Gantry & Terminals	Transmission	2 lot.	152,735	152,735	-	
3	400 kV GIS with Control & Protection Panels with Bus Duct between 400 kV User OHL & GIS.	Connection	2 nos.	2,060,220	-	2,060,220	The full amount is to be paid by the User.
4	400 kV GIS with Control & Protection Panels for Bus Section and Bus Coupler	Common	4 nos.	3,157,328	2,824,978	332,350	B = A*(OETC CB excluding BS&BC CB/Total GIS CB excluding BS&BC CB) B = A*(17/19) C = A*(User CB / Total GIS CB excluding BS&BC CB) C = A*(2/19)
5	11 kV GIS with Control & Protection Panels	Transmission	4 nos.	80,000	80,000	-	The full amount is to be paid by OETC.
6	11 kV GIS with Control & Protection Panels	Connection	2 nos.	40,000	-	40,000	The full amount is to be paid by the User.
7	11 kV GIS with Control & Protection Panels for Bus Section	Common	1 nos.	20,000	13,333	6,667	B = A*(OETC CB excluding BS /Total GIS CB excluding BS) B = A*(4/6) C = A*(User CB / Total GIS CB excluding BS CB) C = A*(2/6)
8	Civil Works for 400 kV Switching Station	Common	lot.	4,079,787	3,725,023	354,764	B = A*(OETC CB /Total GIS CB) B = A*(21/23) C = A*(User CB / Total GIS CB) C = A*(2/23)
9	Earthing and Lightning Protection & Surge Arrestors with Cable Sealing end	Common	lot.	295,996	270,257	25,739	
10	Telecommunication Equipment	Common	lot.	266,213	243,064	23,149	
11	Substation Control System	Common	lot.	378,232	345,342	32,890	
12	Fire Fighting System	Common	lot.	123,459	112,724	10,736	
13	LVAC System & HVAC	Common	lot.	380,359	347,284	33,075	
14	DC System for Grid Station	Common	lot.	290,898	265,602	25,295	
15	EOT Crane for GIS Room	Common	lot.	47,969	43,797	4,171	
16	Security system consisting of CCTV	Common	lot.	411,367	375,596	35,771	
17	Consultancy services charges	Common	lot.	614,730	541,849	72,881	Table 2: Calculation of Consultancy Services for User
18	Calculation of Interest during Construction			232,692	-	232,692	Table 3: Calculation of Interest during Construction for User
	Total Cost / Gross Asset Value (GAV)			**26,023,415**	**22,733,015**	**3,290,400**	

Table 2			
Consultancy Service Charge for OQAE Site Sadaf GS			
Sr. No	Consultancy	C = Total Consultancy Cost	D = Cost for OQAE (Total Consultancy Cost divided by Total Project Cost, and multiplied by Total Connection and Common Assets Value) D = (A/B) *C
1	Design and Supervision	614,730	72,881
Project Cost			
	B = Total Cost [Connection Asset + Common Assets + Transmission Assets] (Table 1)	25,175,993	
	A = Cost for User [Connection Asset + Common Assets] (Table 1)	2,984,827	

Table 3		
Calculation of Interest during Construction		
A = Nominal pre-tax cost of capital	B = Capital Cost including Fixed Asset and Consultancy Charge (GVA table 1)	C = Interest during Construction in OMR = A*B
7.61%	3,057,708	232,692

Table 4					
Connection Charge Offer for User site 400 kV Sadaf GS					
Offers	Years of payment (N)	Nominal pre-tax cost of capital (i)	Annuity factor (Ani) An.i =(1-(1+i)^-N)/i	Annual Connection Charge (CC) to be paid by the User	Total Connection Charge (Capital portion)
One Off Payment (up-front payment) as per Gross Asset Value (GAV) calculated in Table 2)	0	0	0	3,290,400	3,290,400